

Applied Science

(Cambridge OCR AAQ, Extended Certificate)

Do you have an interest in the hands-on application of the Sciences? Do you want to further your skills and knowledge of the three core sciences, Biology, Chemistry and Physics? Are you considering a career in lab work or research?

If the answer to any of these questions is 'yes!' then Applied Science AAQ is the subject for you!

Entry Requirements: In GCSE Combined (Double) Science: grade 4 in both exams or two at grade 4 from any of the three Separate Science GCSEs, and at least a grade 4 in Maths.

Course Content

There are 3 compulsory units on this course which are listed as units 1-3 below. You will complete a further 2 units that are written assignments (NEA) and the school is yet to decide which modules will be chosen so all four options are listed below under units 5/6.

Units	%	When?	Content
Unit 1: Fundamentals of science (Written exam)	20%	Year 12	In this unit you will learn about the key topics that are important in biology, chemistry and physics. You will study two key practicals for each of those components to be assessed as part of section D of the exam. Topics include: Cell structure and microscopy - Bioenergetics - Structure and function of biological molecules - Biodiversity and ecosystems - Atomic structure and the Periodic Table - Quantitative chemistry - Structure and bonding - Rates of reactions and enthalpy changes - Electricity - Motion - Medical physics - Practicals
Unit 2: Investigating Science (NEA)	20%	Year 12	In this unit you will learn about the role of a research scientist in industry by learning how to conduct your own scientific investigation. You will develop the skills to research, plan and risk assess your investigation before safely undertaking the practical tasks. Topics include: - Planning a scientific investigation - Performing a scientific investigation - Analysing and communicating results - Evaluating a scientific investigation
Unit 3: Science in society (written exam)	20%	Year 13	In this unit you will learn about the skills scientists use and the roles they perform in an international scientific community. You will examine different types of scientific data and learn how scientists use them to draw conclusions that can contribute to scientific advancement. Topics include: - What scientists do - Handling scientific data - Scientific developments - Communicating science
Units 5/6 (NEA)	40%		Analytical techniques in chemistry OR Environmental studies OR Forensic biology OR Medical physics

Career Links

This qualification is supported by a range of universities and taken alongside other qualifications it can fulfil the entry requirements for a number of science-related higher education courses, including biomedical, forensic and sports science, as well as nursing. This qualification could also support learners in progressing to a related apprenticeship or into employment.

For more information please contact Mr Lake